5 Oscillations at low polynomial degree for unsteady flow through rectangular channel

...Not yet complete

6 Analysis of accuracy of upwind PHS-RBF scheme for unsteady advection-diffusion equation

We shall study the accuracy of the scheme developed herein. In order to do so, we shall simulate the temperature field for a constant-velocity travelling wave in a diffusive medium. The analytical expression for such a temperature field is given as

$$T(x, y, t) = \exp(-\alpha k^2 t) \sin(k_1 x + k_2 y - \omega t) \tag{8}$$

where $k_1 = k \cos\theta$, $k_2 = k \sin\theta$, $\omega = k \|\boldsymbol{u}\|$, where k is the wavenumber, $\|\boldsymbol{u}\| = \sqrt{u^2 + v^2}$, $\theta = \tan^{-1} v/u$.
This gives an initial condition of

$$T(x, y, 0) = \sin(k_1 x + k_2 y) \tag{9}$$

Thus, we use the upwind PHS-RBF scheme developed here to numerically solve equation (10) with the above initial condition, and compute the errors from the analytical travelling wave solution.

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \alpha\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) \tag{10}$$

We first solve the diagonally moving travelling wave with no diffusion. The variation of the L2 norm of the errors with the solution time, and the h-refinement of the maximum value of these errors to obtain the order and rate of convergence, are shown in the figures below (9, 10). From the figures below, we observe that the error is initially zero, then steadily rises and then saturates. It then shows small oscillations, but overall, it remains bounded at this value. The oscillations are a result of the dispersive errors in the scheme, which cause phase lags between the analytical solution and the numerical solution. From the figure 10, we also observe that, although $f_u = 1.0$ has a higher value of error at a coarse grid, it also shows nearly second-order convergence to zero as the grid is refined. As we decrease f_u, the order of convergence also decreases, and the method becomes closer to first-order accurate. We have restricted the polynomial degree to 2, since using higher polynomial degrees causes oscillations downstream of any sharp discontinuity. In order to obtain higher-order accurate upwinding, some modification in the method, perhaps something similar to TVD schemes is needed. The order and rate of convergence for various upwinding fractions are given in the table 5.